SORTIE: Fair Vehicle Routing When Route Durations Are Random.

Ex-post ordered fairness, rank prices and risk-consistent tours

Quang-Vinh Dang
British University Vietnam, Hung Yen, Vietnam
vinh.dq4@buv.edu.vn ORCID 0000-0002-3877-8024

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Abstract

Fair vehicle routing balances drivers' workloads with ordered measures of route lengths, such as the range or the Gini mean difference. The recent exact approaches, which also require every route to be a shortest tour of its customers, assume deterministic travel times. Drivers, however, experience realised durations. We study fair routing when arc travel times are random, with a fleet-wide day factor and arc-level noise. We distinguish ex-ante fairness (of expected workloads) from ex-post fairness (the expected unfairness of the realised day, sorted day by day) and show that every deterministic fair plan overstates ex-post fairness, with an explicit bound in terms of the routes' quadratic lengths. We prove that multiplicative common shocks leave expected ordered fairness unchanged. We then give an exact, sampling-free formula for the expected value of any ordered measure of independent route durations, a Poisson–binomial order-statistic integral that also handles the negative weights of dispersion measures, and derive from it *rank prices*: the derivative of expected unfairness with respect to a shift of each route's duration. Negative rank prices identify exactly the routes a fairness objective would like to lengthen. Under uncertainty even the min-max measure can prefer a tour that is not the shortest, so route consistency must be imposed explicitly; we define it through each driver's own mean–standard deviation and prove that such tours are supported Pareto tours, computable by parametric Held–Karp. The resulting bilevel heuristic, SORTIE, screens all relocate and swap moves with the rank prices. On instances derived from the CVRPLIB benchmark used by the deterministic literature, deterministic fair plans understate their realised unfairness by 8–27%, while the integral agrees with Monte Carlo to about 0.1% at under a millisecond per plan. Optimising ex-post fairness improves realised fairness by about 1% when noise is proportional to length and by about 1.5% under congestion-driven heterogeneous noise; screening with rank prices gives plans 6–9% better than random screening; and without the lower level, nearly all range-optimised plans buy fairness with detours.

Keywords: vehicle routing; fairness; stochastic travel times; ordered weighted averaging; order statistics; bilevel optimisation.

1 Introduction

Routing plans that minimise total distance often give some drivers much longer days than others. Unequal workloads hurt driver satisfaction and retention and can be seen as unfair [Matl et al., 2018]. The fairness literature in routing therefore optimises a *fairness measure* of the vector of route workloads: the longest route, the range, the Gini mean difference (GMD), the variance or a

mean absolute deviation [Matl et al., 2018, Karsu and Morton, 2015, Bektaş and Letchford, 2020]. Ordered measures, which weight the sorted workloads [Yager, 1988, Ogryczak and Śliwiński, 2003, Nickel and Puerto, 2005], cover all of these in one framework [Blanco et al., 2026].

Non-monotone measures such as the range or the GMD create a well-known pathology: an optimiser can lower the measure by making a light route *longer*, giving a driver a needless detour [Matl et al., 2018]. van Rossum et al. [2026a,b] and Ljubić et al. [2026] close this loophole exactly, by requiring every route to be a shortest (TSP-optimal) tour of its customers. Ljubić et al. [2026] write the problem as a bilevel programme in which a planner assigns customers and each driver solves a TSP, and solve instances with up to 35 customers to optimality for seven measures.

All of these works assume deterministic travel times. A driver's day, however, is a realised duration: congestion, weather, parking and service-time variability differ between routes and between days. Fairness is then a statement about random variables, and three questions arise that the deterministic theory does not ask.

1. *What should be balanced?* The expected workloads (ex-ante fairness), or the workloads drivers actually experience, sorted day by day (ex-post fairness)? The identity of the worst-off driver changes from day to day, and the two notions can disagree.

2. *How can ex-post fairness be optimised?* Sample-average approximation (SAA) is the default, but sorting per scenario makes the objective noisy and its gradient uninformative.

3. *What is a consistent route?* Under random durations "the shortest tour" is not obviously what a driver would choose, and we show that monotone measures no longer protect drivers from detours.

Contributions.

- **Model.** We formulate fair routing with random travel times (a fleet-wide day factor times arc-level noise) as a bilevel problem with an ex-post ordered fairness objective, an expected-cost budget and per-driver risk-consistent tours. With no noise it reduces to the X-CVRP-TSP of Ljubić et al. [2026].

- **Fairness illusion (Proposition 5).** For every ordered measure with nondecreasing weights, ex-ante fairness underestimates ex-post unfairness, by at most $\|\lambda\|_1 v(\sum_k W_k)^{1/2}$, where W_k is the route's quadratic length. Common multiplicative shocks leave the expected value of every ordered measure unchanged (Proposition 2); they do inflate the variance measure and the tail of unfairness.

- **PBOI (Theorem 3).** An exact, sampling-free formula for the expected ordered measure of independent route durations with given marginals (we use moment-matched gamma marginals), as an integral of a Poisson–binomial distribution against tail sums of the weights. It handles negative weights and costs $O(Gm^2)$.

- **Rank prices (Proposition 4).** The derivative of expected unfairness with respect to a location shift of a route's duration equals the expected weight of its rank. Prices of dispersion measures sum to zero, and a negative price is precisely the stochastic incentive to lengthen a route.

- **Risk-consistent tours (Proposition 6, Theorem 8).** Even $\mathbb{E}\max$ can prefer a non-shortest tour. We define consistency through the driver's own mean–standard deviation and

prove that the consistent tour is a supported Pareto tour of (length, quadratic length), found exactly by dichotomic parametric Held–Karp.

- **Algorithm and evidence.** SORTIE, a bilevel local search whose moves are screened by rank prices, is compared with the deterministic fair plan, SAA and ablations on the instances used by the deterministic literature, with exact enumeration on small instances.

2 Literature review

Workload equity in deterministic routing. Matl et al. [2018] survey equity measures for routing workloads and show that their monotonicity governs whether optimal plans can contain needlessly inefficient routes; Matl et al. [2019a] study how the choice of workload resource changes balanced plans, and Matl et al. [2019b] approximate cost–balance Pareto fronts. Min-max routing has a long exact tradition [Applegate et al., 2002, Bertazzi et al., 2015], and bi-objective route balancing is solved exactly by Halvorsen-Weare and Savelsbergh [2016]; ℓ_p norms as fairness measures appear in Bektaş and Letchford [2020]. Recent work on non-monotone and ordered objectives includes branching rules for range and order-based objectives [van Rossum et al., 2024, 2026a], cutting planes that enforce TSP-optimality for the range [van Rossum et al., 2026b], route-cost equity constraints [Agius et al., 2026], Gini-bounded mTSP variants [Bhadoriya et al., 2026], fair cooperative fleets [López Sánchez et al., 2024], driver satisfaction [Bruinink et al., 2025], multi-period equity [Nekooghadirli et al., 2026] and workload/pay design for heterogeneous or crowdsourced drivers [Mahmoodian et al., 2025, Sobhanan et al., 2026]. The ordered machinery comes from location theory [Nickel and Puerto, 2005, Ogryczak, 2000, Blanco et al., 2024, Ljubić et al., 2024, Blanco et al., 2026], the ordered median TSP [Ljubić et al., 2026], OWA [Yager, 1988, Ogryczak and Śliwiński, 2003] and top-k sums [Ogryczak and Tamir, 2003]. Closest to us, Ljubić et al. [2026] propose a unified ordered framework with per-vehicle TSP-optimality through bilevel optimisation. All of these works are deterministic.

Fairness under uncertainty. In routing, Kinable et al. [2026] study a stochastic VRP with partial deliveries where fairness is among *customers* (expected fill rates) under uncertain demand, and Gao et al. [2026] combine sequential fair allocation and routing with ex-post max-min objectives across locations. Consistency in the sense of the consistent VRP (same driver, similar arrival times) is treated under stochastic customers and demands by Alvarez et al. [2024] and with robust demand by Wu et al. [2025], following Groër et al. [2009], Kovacs et al. [2015], Yu et al. [2024]; this notion differs from the TSP-optimality consistency studied here. For drivers, Anh [2022] bound workload differences in task assignment with a distributionally robust chance constraint and Lin et al. [2022] add robust workload fairness to two-echelon meal-delivery routing; gig and rideshare work considers fairness of assignment probabilities and matching under stochastic arrivals [Singh et al., 2023, Basik et al., 2021, Nanda et al., 2020, Hosseini and Schierreich, 2025]. The ex-ante/ex-post distinction is established in welfare economics [Fleurbaey and Peragine, 2012], in inequity-averse stochastic optimisation [Gutjahr, 2021], in Gini-based stochastic shelter location and humanitarian logistics [Mostajabdaveh et al., 2019, Alem et al., 2022, Mostajabdaveh et al., 2025] and in sequential allocation [Sinclair et al., 2023]; see also Shehadeh [2023], Tsang and Shehadeh [2026], Ye et al. [2024], Barak and Talgam-Cohen [2026]. To our knowledge no paper optimises an ordered measure of *random route durations*, separates its ex-ante and ex-post versions, or defines route consistency for random travel times.

Stochastic travel times. A-priori routing with random travel times goes back to Laporte et al. [1992] and Kenyon and Morton [2003], the latter with makespan-type objectives; see Gómez et al. [2016], Adulyasak and Jaillet [2016], Errico et al. [2016, 2018], Zhang et al. [2013], Ehmke et al. [2015], Osorio-Mora et al. [2025] and the surveys Oyola et al. [2018], Gendreau et al. [2016]. Correlated travel times are handled by Rostami et al. [2021], Bakach et al. [2021], Rajabi-Bahaabadi et al. [2019], Bomboi et al. [2021], robust and distributionally robust variants by Han et al. [2014], Jaillet et al. [2016], Zhang et al. [2021], Ghosal and Wiesemann [2020], and SAA by Verweij et al. [2003]. Mean–risk path choice [Nikolova and Stier-Moses, 2014, Sen et al., 2001] motivates our driver-level consistency criterion. CVaR [Rockafellar and Uryasev, 2000, 2002] and the price of fairness [Bertsimas et al., 2011, Mo and Walrand, 2000] complete the background.

Order statistics and parametric optimisation. Order statistics of independent, non-identically distributed variables have permanent representations [Vaughan and Venables, 1972, Bapat and Beg, 1989, Balakrishnan, 2007, Glueck et al., 2008] and are related to the Poisson–binomial count of successes in independent trials [Hoeffding, 1956, Fernandez and Williams, 2010, Hong, 2013, Biscarri et al., 2018]; see David and Nagaraja [2003], Sen [1970]. Supported efficient solutions of bi-criteria problems are found by dichotomic search [Aneja and Nair, 1979, Ehrgott, 2005, Ehrgott and Gandibleux, 2000, Ralphs et al., 2006], and the TSP on few nodes by the Held–Karp recursion [Held and Karp, 1962, Bellman, 1962].

3 Model

Let $G = (V, A)$ be a complete graph with depot 0, customers $V' = \{1, \dots, n\}$, symmetric costs c_{ij} satisfying the triangle inequality, demands q_i and m identical vehicles of capacity Q. A plan consists of a partition $S_1, \dots, S_m$ of V' into nonempty capacity-feasible sets and a tour r_k of $\{0\} \cup S_k$ for each vehicle.

Random workloads. The travel time of arc (i, j) is $\tau_{ij} = Y c_{ij} \varepsilon_{ij}$, where $Y = \exp(s_Y Z - s_Y^2/2)$, $Z \sim N(0, 1)$, is a fleet-wide day factor with $\mathbb{E}Y = 1$, and the arc disturbances ε_{ij} are independent of each other and of Y with mean 1 and variance v^2 (gamma in our experiments). The workload of driver k is $\Phi_k = Y X_k$ with

$$X_k = \sum_{(i,j)\in r_k} c_{ij}\varepsilon_{ij}, \qquad \mathbb{E}X_k = L_k = \sum_{(i,j)\in r_k} c_{ij}, \qquad \operatorname{Var} X_k = \sigma_k^2 = v^2 W_k, \quad W_k = \sum_{(i,j)\in r_k} c_{ij}^2.$$

L_k is the tour length and W_k its *quadratic length*. Given Y the workloads are independent; unconditionally they are positively correlated.

Ordered fairness. For $a \in \mathbb{R}^m$ with order statistics $a_{(1)} \le \cdots \le a_{(m)}$ and weights $\lambda \in \mathbb{R}^m$, $\ell_\lambda(a) = \sum_j \lambda_j a_{(j)}$. We use the maximum ($\lambda = e_m$), the range ($e_m - e_1$), the GMD ($\lambda_j = 2(2j - m - 1)/m^2$) and the mean absolute deviation from the median (MAD; $\lambda = (-1, \dots, -1, 0, 1, \dots, 1)/m$ for odd m), together with the empirical variance $\operatorname{Var}_{\text{emp}}(a) = m^{-1} \sum_k (a_k - \bar{a})^2$. All four λ are nondecreasing.

Definition 1 (Three notions of fairness)**.** For a plan with random workloads Φ: *ex-ante* unfairness $\ell_\lambda(\mathbb{E}\Phi)$; *ex-post* unfairness $\mathbb{E}\ell_\lambda(\Phi)$; *risk of unfairness* $\operatorname{CVaR}_\beta \ell_\lambda(\Phi)$.

A deterministic fair model optimises the ex-ante notion. Drivers experience the ex-post one. SORTIE optimises ex-post unfairness and reports the CVaR out of sample.

4 Theory

Proposition 2 (Common shocks)**.** *For every λ, $\mathbb{E}\ell_\lambda(\Phi) = \mathbb{E}\ell_\lambda(X)$, and $\mathbb{E}\,\mathrm{Var}_{\mathrm{emp}}(\Phi) = e^{s_Y^2}\,\mathbb{E}\,\mathrm{Var}_{\mathrm{emp}}(X)$.*

Proof. Sorting commutes with multiplication by $y > 0$, so $\ell_\lambda(yX) = y\ell_\lambda(X)$; take expectations using independence of Y and X and $\mathbb{E}Y = 1$. The empirical variance is homogeneous of degree two and $\mathbb{E}Y^2 = e^{s_Y^2}$. □

Fleet-wide shocks therefore drop out of the planning objective for ordered measures, which reduces evaluation to the conditionally independent vector X.

Theorem 3 (Poisson–binomial order-statistic integral)**.** *Let $X_1, \dots, X_m \geq 0$ be independent with CDFs F_k. For $t \geq 0$ let $N(t) = \#\{k : X_k \leq t\}$, a Poisson–binomial count with success probabilities $F_k(t)$, and $\pi_r(t) = \mathbb{P}(N(t) = r)$. With tail sums $\Lambda_r = \sum_{j=r+1}^{m} \lambda_j$,*

$$\mathbb{E}\ell_\lambda(X) = \int_0^\infty \sum_{r=0}^{m-1} \Lambda_r\, \pi_r(t)\, dt. \tag{1}$$

Proof. $X_{(j)} > t$ if and only if fewer than j of the X_k are at most t, so $\mathbb{E}X_{(j)} = \int_0^\infty \mathbb{P}(N(t) \leq j-1)\, dt$. Hence $\mathbb{E}\ell_\lambda(X) = \int_0^\infty \sum_j \lambda_j \sum_{r \leq j-1} \pi_r(t)\, dt = \int_0^\infty \sum_r \pi_r(t) \sum_{j \geq r+1} \lambda_j\, dt$, where Tonelli's theorem is applied to the positive and negative parts of λ separately. □

The vector $\pi(t)$ is the coefficient vector of $\prod_k (1 - F_k(t) + F_k(t)z)$, obtained by m convolution steps. Below the smallest lower quantile the integrand equals Λ_0, above the largest upper quantile it vanishes ($\Lambda_m := 0$), and in between we use Gauss–Legendre panels whose breakpoints are quantiles of *every* marginal, so that a short, low-variance route is resolved even when the other routes are long. We use moment-matched gamma marginals $X_k \sim \Gamma(L_k^2/\sigma_k^2, \sigma_k^2/L_k)$. The identity behind (1) is classical for order statistics of non-identical variables [Bapat and Beg, 1989, Balakrishnan, 2007]; our contribution is to make it an exact planning objective for ordered fairness, where the tail sums Λ_r absorb the negative weights of dispersion measures, and to differentiate it.

Proposition 4 (Rank prices)**.** *Let R_k be the rank of X_k and define $\varpi_k = \sum_j \lambda_j \mathbb{P}(R_k = j)$, where $\mathbb{P}(R_k = j) = \int f_k(t)\, \mathbb{P}(N_{-k}(t) = j-1)\, dt$ and N_{-k} counts the other drivers. Then $\frac{d}{d\delta}\mathbb{E}\ell_\lambda(X + \delta e_k)\big|_{\delta=0} = \varpi_k$ and $\sum_k \varpi_k = \sum_j \lambda_j$.*

Proof. ℓ_λ is piecewise linear with $\partial\ell_\lambda/\partial a_k = \lambda_{R_k(a)}$ off the null set of ties and is $\|\lambda\|_1$-Lipschitz, so differentiation and expectation commute by dominated convergence. The ranks form a permutation, so $\sum_k \mathbb{P}(R_k = j) = 1$. □

For the range, GMD and MAD the prices sum to zero: burdened drivers carry positive prices and light drivers negative ones. A negative price is exactly the incentive to lengthen a route. Deterministically, the price of the lightest route under the range is -1, and a detour of length δ reduces the range by δ; stochastically, the incentive is spread over all drivers that might be the lightest on a given day. SORTIE uses the prices, together with $\psi_k = \partial\mathbb{E}\ell_\lambda/\partial\sigma_k$, to predict the effect of a move.

Proposition 5 (Fairness illusion)**.** *If λ is nondecreasing then ℓ_λ is convex and*

$$0 \leq \mathbb{E}\ell_\lambda(X) - \ell_\lambda(L) \leq \|\lambda\|_1\, \mathbb{E}\|X - L\|_\infty \leq \|\lambda\|_1\, v\Big(\sum_k W_k\Big)^{1/2}.$$

Moreover $\mathbb{E}\,\mathrm{Var}_{\mathrm{emp}}(X) = \mathrm{Var}_{\mathrm{emp}}(L) + \frac{m-1}{m^2}\sum_k \sigma_k^2$, and for Gaussian marginals $\mathbb{E}\,\mathrm{GMD}(X) = m^{-2}\sum_{i \neq j}\left[s_{ij}\sqrt{2/\pi}\, e^{-d_{ij}^2/(2s_{ij}^2)} + d_{ij}(1 - 2\Phi(-d_{ij}/s_{ij}))\right]$ with $d_{ij} = L_i - L_j$ and $s_{ij}^2 = \sigma_i^2 + \sigma_j^2$.

Proof. For nondecreasing λ the rearrangement inequality gives $\ell_\lambda(a) = \max_\sigma \sum_j \lambda_j a_{\sigma(j)}$, a maximum of linear functions; Jensen's inequality gives the lower bound. Since $|a_{(j)} - b_{(j)}| \le \|a - b\|_\infty$ for every j, ℓ_λ is $\|\lambda\|_1$-Lipschitz in the sup norm, and $\mathbb{E}\max_k |Z_k| \le (\mathbb{E}\sum_k Z_k^2)^{1/2}$. The variance identity follows from $\mathbb{E}X_k^2 = L_k^2 + \sigma_k^2$ and $\mathbb{E}\bar{X}^2 = \bar{L}^2 + m^{-2}\sum_k \sigma_k^2$; the GMD formula is the mean of a folded normal. □

A deterministic fair plan therefore *overstates* the fairness drivers will experience. The overstatement grows with the arc noise and with the quadratic lengths: long arcs, not long routes, are what make a balanced plan unbalanced in practice.

Proposition 6 (Monotonicity no longer protects drivers)**.** *With deterministic workloads, every monotone measure ($\lambda \ge 0$) has an optimal plan made of TSP-optimal routes [Matl et al., 2018]. With random workloads this fails even for* $\mathbb{E}\max$.

Example 7. Two drivers; $X_1 \equiv 100$. Driver 2 can drive tour A ($L = 100$, $\sigma = 10$) or tour B ($L = 101$, $\sigma \approx 0$). With Gaussian X_2, $\mathbb{E}\max = 100 + 10\varphi(0) \approx 103.99$ under A and ≈ 101 under B, so the optimiser picks the longer tour B.

Route consistency must therefore be imposed for every measure, and it needs a risk attitude. We let each driver rank the tours of their customer set by

$$\rho_\kappa(r) = \frac{\mathbb{E}[\Phi(r) \mid Y] + \kappa\,\mathrm{SD}[\Phi(r) \mid Y]}{Y} = L(r) + \kappa v\sqrt{W(r)},$$

which ranks tours identically on every day. $\kappa = 0$ is TSP-optimality (ties broken by smaller W).

Theorem 8 (Risk-consistent tours are supported)**.** *For $\kappa \ge 0$, some minimiser of ρ_κ over the tours of a fixed customer set is a supported extreme tour: it minimises $L(r) + \theta W(r)$ for some $\theta \in [0, \infty)$, or minimises W lexicographically after L ($\theta = \infty$). All supported extreme tours are enumerated by dichotomic search, each step being one Held–Karp run with arc weights $c_{ij} + \theta c_{ij}^2$.*

Proof. $f(L, W) = L + \kappa v\sqrt{W}$ is concave and nondecreasing on $\mathbb{R}^2_+$. A concave function bounded below on a pointed polyhedron attains its minimum at an extreme point; since f is nondecreasing, the minimum of f over the finite set $P = \{(L(r), W(r))\}$ equals its minimum over $\mathrm{conv}(P) + \mathbb{R}^2_+$ and is attained at an extreme point of that set. These extreme points are exactly the unique minimisers of $w_1 L + w_2 W$ with $(w_1, w_2) \ge 0$, $(w_1, w_2) \ne 0$ (for $w_1 = 0$ with ties broken by smaller L), and dichotomic search enumerates all of them [Aneja and Nair, 1979]. Tours in the relative interior of a hull edge are dominated by the edge's end points and need not be enumerated. With heterogeneous noise (ε_{ij} with variance $v^2 H_{ij}^2$) the same holds with $W = \sum (H_{ij} c_{ij})^2$ and arc weights $c_{ij} + \theta H_{ij}^2 c_{ij}^2$. □

The SORTIE problem.

$$\begin{aligned} \min\ & \mathbb{E}\,\ell_\lambda\big(X(r_1, \dots, r_m)\big) \\ \text{s.t.}\ & (S_1, \dots, S_m) \text{ capacity-feasible partition},\ S_k \ne \emptyset, \qquad \textstyle\sum_k L(r_k) \le B \cdot C^*, \\ & r_k \in \arg\min\{\rho_\kappa(r) : r \text{ a tour of } \{0\} \cup S_k\} \quad \forall k, \end{aligned}$$

where C^* is the minimum expected total length and $B \ge 1$ the budget factor. With $v = 0$ and $\kappa = 0$ this is the X-CVRP-TSP of Ljubić et al. [2026].

Algorithm 1 SORTIE

$P \leftarrow$ cost-efficient plan; $P^* \leftarrow P$
while time remains **do**
    compute rank prices ϖ_k and dispersion prices ψ_k of P (Prop. 4)
    score every relocate and swap move by $\sum_{k\in\{a,b\}} \varpi_k \Delta L_k + \psi_k \Delta\sigma_k$ (insertion deltas)
    evaluate the K best-scored budget- and capacity-feasible moves with screening tours and PBOI (Thm. 3)
    confirm the improving candidates in order with the exact lower level (Thm. 8)
    if a confirmed move improves P **then** apply it
    else ruin (price-weighted or related removal, at most 12 customers) and recreate (price-scored regret-2); repair cost if over budget; confirm every changed route; accept by simulated annealing
    end if
    update P^*
end while
return P^*

5 The SORTIE algorithm

The lower level is exact for routes of up to 15 customers (numba Held–Karp, memoised by customer set, about 5 ms at 15 customers) and uses 2-opt/Or-opt from several starts beyond that, including a warm start built from the route's previous tour by cheapest insertion. Solving it for every candidate move would dominate the run time, so candidates are evaluated with *screening tours*: a single warm-started 2-opt/Or-opt descent (for $\kappa > 0$, inside the same dichotomic search). A move is applied only after both changed routes have been re-solved by the exact lower level and the move still improves the objective and respects the budget; perturbed plans are confirmed route by route in the same way. Every plan the search accepts therefore satisfies the bilevel constraint; the screening tours only decide which moves are examined. Because a screening tour is never shorter than the exact one, a move that improves only with exact tours can be missed; this costs search quality, not feasibility. The evaluator costs about 0.4 ms per plan for $m = 5$. Together with a cap of 12 customers on the ruin size, this raises the iteration rate of SORTIE-0 at $n = 100$ from about 250 to about 1700 per minute, and at $n = 75$ from about 170 to about 1800.

6 Computational study

6.1 Set-up

Instances. We follow the slicing rule of the deterministic fair routing literature [Ljubić et al., 2026]. Block b of size n consists of the depot and the n consecutive customers $bn+1, \ldots, (b+1)n$ of CVRPLIB X-n641-k35 [Uchoa et al., 2017], with ceiled Euclidean distances, $m = 5$ vehicles and capacity $Q = \lfloor \sum_i q_i/4 \rfloor$, so one vehicle's worth of slack is available. The main grid uses $n \in \{15, 20, 25, 30, 35\}$ and blocks $b = 0, \ldots, 4$ (25 instances); the scale test uses $n \in \{50, 75, 100\}$ and blocks 0–5; the exactness test uses ten instances with $n = 10$, $m = 3$ and $Q = \lfloor \sum_i q_i/2.4 \rfloor$.

Uncertainty. Unless stated otherwise $v = 0.3$ (gamma arc noise) and $s_Y = 0.15$. In the congestion experiment (E8) the arc coefficient of variation is vH_{ij}, where $H_{ij} \in [1/3, 2]$ is largest for arcs near a randomly drawn congestion centre, so noise is no longer proportional to length.

Table 1: E1: PBOI against arc-level Monte Carlo (50 000 samples) on 168 plans; z: difference over its Monte Carlo standard error.

Measure	Plans	Mean |err| (%)	Max |err| (%)	Share $\|z\| < 3$
Max	168	0.041	0.137	0.994
Range	168	0.062	0.249	1.000
GMD	168	0.057	0.241	1.000
MAD	168	0.064	0.272	0.988
Var	168	0.108	0.440	0.994

Budget. C^* is the cost of the best of three restarts of the cost-minimising search; the total length of every fair plan must satisfy $\sum_k L_k \le B\, C^*$ with $B \in \{1.05, 1.10\}$.

Methods. All methods share the search of Algorithm 1, the start plan, the time limit (15 s for $n \le 25$, 25 s for $n \ge 30$, 120 s for $n \ge 50$) and the random seed. They differ only in the objective and in the lower level. *Det* minimises the deterministic ordered measure $\ell_\lambda(L)$ of the tour lengths, i.e. the model of Ljubić et al. [2026] solved heuristically. *SAA* minimises the sample average of $\ell_\lambda(X)$ over 200 fixed arc scenarios. *SORTIE-0* minimises the exact ex-post value by PBOI with TSP-optimal tours ($\kappa = 0$); *SORTIE-1* uses the risk-consistent lower level with $\kappa = 1$. In Det and SAA the moves are screened by the deterministic and the sample objective respectively.

Evaluation. Every plan is evaluated out of sample by arc-level Monte Carlo with 20 000 scenarios (10 000 for $n \ge 50$), including the day factor, with common random numbers across methods. The reported *ex-post value* is the Monte Carlo estimate of $\mathbb{E}\ell_\lambda(\Phi)$. Paired comparisons use the Wilcoxon signed-rank test. All tables are generated from the raw results by the released scripts.

6.2 Accuracy and cost of the evaluator (E1, E2)

Table 1 compares PBOI with 50 000-sample arc-level Monte Carlo on 168 plans: the cost-efficient plans and random feasible plans whose routes are sequenced by the lower level, for $n \in \{15, 25, 35, 50\}$. The mean absolute relative error is below 0.11% for all five measures and below 0.44% in the worst case, and 99–100% of the differences lie within three Monte Carlo standard errors. The mean z-scores are not exactly zero (up to 0.34 for the MAD): PBOI carries a systematic bias of a few hundredths of a percent, which we attribute to the moment-matched gamma marginal of each route (a sum of arc gammas with different scales is not exactly gamma). PBOI takes 0.44 ms per plan for $m = 5$, against 0.28 s for 20 000-scenario Monte Carlo. A 200-scenario SAA is about seven times faster than PBOI but is a noisy estimate. The advantage of PBOI is therefore that it is exact and noise-free at a cost compatible with thousands of evaluations per second, not raw speed.

On the ten $n = 10$ instances we enumerated all capacity- and budget-feasible partitions (4 to 145 per instance) with exact Held–Karp tours. The cost search found the optimal C^* on all ten. For both range and GMD, SORTIE-0 found the plan with the smallest exact ex-post value on nine of ten instances (mean gap 0.39%, maximum 3.9%). Det found the deterministic optimum on nine of ten; on the tenth instance, which has only 20 budget-feasible partitions, it stopped at a plan with twice the optimal deterministic range.

Table 2: E3: what each method believes its unfairness is, relative to the true ex-post value (Monte Carlo without the day factor, which is what each model represents), mean signed error over 50 runs per cell. Last column: largest absolute error of the PBOI prediction for any method's plan.

	Believed vs true (mean signed %)				PBOI
Measure	Det	SAA	SORTIE-0	SORTIE-1	max |err| %
Max	-7.97	-0.71	-0.01	-0.01	0.20
Range	-17.56	-0.99	+0.03	+0.02	0.70
GMD	-14.68	-0.91	+0.04	+0.05	0.61
MAD	-16.17	-0.88	+0.05	+0.07	0.68
Var	-27.23	-1.33	+0.00	+0.02	0.98

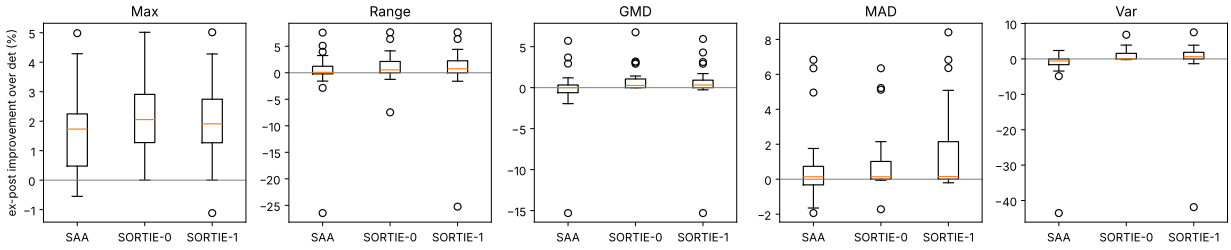


Figure 1: E3, $B = 1.10$: per-instance improvement of the ex-post value over Det.

6.3 The fairness illusion and the value of ex-post optimisation (E3)

Table 2 quantifies Proposition 5. A deterministic fair plan believes its unfairness is the ex-ante value; the truth is 8% larger for the maximum, 15–18% larger for the range, GMD and MAD, and 27% larger for the variance. The in-sample SAA value is optimistic by about 1%, which is the optimiser's curse on 200 scenarios. The PBOI value believed by SORTIE has a mean signed error below 0.1%, a mean absolute error of 0.07–0.23% and a worst case of 1%.

Table 3 reports the out-of-sample ex-post value relative to Det. The signed-rank test favours SORTIE-0 and SORTIE-1 over Det in 9 of the 10 cells each. Pooled over all 250 runs, SORTIE-0 improves on Det by 0.89% on average (median 0.23%; better in 141 runs, tied in 96, worse in 13; $p < 10^{-18}$) and SORTIE-1 by 0.77% (better in 154, worse in 31). SAA improves by only 0.14% (median zero; better in 115, worse in 74) and is favoured significantly in 3 cells. Head to head, SORTIE-0 beats SAA in 145 runs and loses in 15, and SORTIE-1 beats it in 147 and loses in 27 ($p < 10^{-17}$). The largest gains are for the maximum at $B = 1.10$ (2.1%, 24 of 25 instances), where the extra budget lets the planner trade a slightly longer but less variable longest route.

The improvements are small, about one percent on average, and this is itself a finding. With homogeneous noise ($\sigma_k^2 = v^2 W_k$) the standard deviation of a route grows with its length, so the plans that balance lengths also nearly balance risks, and a deterministic fair plan is close to ex-post optimal even though its *assessment* of its own fairness is badly wrong. For context, the cost-optimal plan has an ex-post range of 2087 against about 1340 for the fair plans, at a cost 6.5% higher: fairness itself is worth optimising; the stochastic refinement adds about one percent on top. For range-optimised plans the $\mathrm{CVaR}_{0.9}$ of the realised range, which none of the methods optimises, is 2054 for Det, 2059 for SAA, 2037 for SORTIE-0 and 2032 for SORTIE-1; both SORTIE variants improve it at both budgets, SAA at neither.

Table 3: E3: improvement of the out-of-sample ex-post value over Det (mean %, positive is better; in brackets the number of instances on which the method is strictly better). *: Wilcoxon signed-rank $p < 0.05$ in favour of the method; the test is on ranks and can disagree in sign with the mean when one instance dominates (Var, $B = 1.10$, SORTIE-1).

Measure	B	SAA	SORTIE-0	SORTIE-1
GMD	1.05	-0.40 (6/25)	-0.09 (8/25)	-0.03 (9/25)
GMD	1.10	-0.32 (9/25)	+0.94* (14/25)	+0.34* (17/25)
MAD	1.05	+0.86* (11/25)	+1.10* (13/25)	+1.24* (12/25)
MAD	1.10	+0.70 (13/25)	+0.96* (14/25)	+1.53* (18/25)
Max	1.05	+0.71* (14/25)	+0.61* (15/25)	+0.77* (15/25)
Max	1.10	+1.70* (23/25)	+2.08* (24/25)	+1.93* (24/25)
Range	1.05	+0.12 (13/25)	+0.64* (15/25)	+1.56* (15/25)
Range	1.10	-0.21 (13/25)	+1.07* (17/25)	+0.39* (15/25)
Var	1.05	+0.48 (7/25)	+0.50* (9/25)	+0.48* (13/25)
Var	1.10	-2.25 (6/25)	+1.09* (12/25)	-0.49* (16/25)

6.4 Route consistency (E4)

E4 removes the lower level: the planner may sequence each route freely (*free*) instead of using the TSP-optimal tour (*lower*). For the maximum, free sequencing changes almost nothing, as monotonicity predicts in the deterministic case. For the range and the GMD the free planner lengthens the short routes: 23 of 25 Det plans and 24 of 25 SORTIE plans for the range contain a route that is not a shortest tour of its customers, by 88–108 distance units on average. This lowers the ex-post value by 1–3% for Det on both measures and for SORTIE on the range (Table 4); for SORTIE on the GMD it is 0.9% worse, because the larger space slows the search. Wherever it helps, the reduction comes from making some drivers drive detours. This is exactly the behaviour signalled by negative rank prices and excluded by the bilevel constraint. The deterministic belief error persists with free tours (-24% for the range); the PBOI belief remains exact.

6.5 Rank-price screening (E5)

Replacing the price screen by $K = 8$ random feasible moves, with everything else unchanged, gives a similar number of iterations but worse plans: price screening lowers the ex-post value by 5.6% for the range and 8.5% for the GMD on average (medians 0% and 5.4%; better on 12 and 17 of 25 instances, worse on 3 and 2, ties otherwise; $p = 0.005$ and $p < 10^{-3}$). For the range the mean is driven by a few instances on which random screening stalls far from a good plan; for the GMD the advantage is broad. The rank prices are thus an effective first-order guide for the search, and not only an interpretive device.

6.6 Noise level (E6)

Table 5 varies v for $n = 25$. The illusion grows monotonically with v, as the bound of Proposition 5 suggests: at $v = 0.5$ a deterministic plan understates its range by 43%. The gain of SORTIE over Det also grows with v for the range, from -0.5% at $v = 0.1$ (where the illusion is small and the differences are search noise) to 2.1% at $v = 0.5$ (better on all five instances); for the GMD it is 1.1%, 0.7% and 1.8%.

Table 4: E4: free tours against lower-level tours, $B = 1.10$, 25 instances per row. [a]Runs with at least one route that is not a shortest tour of its customers. [b]Mean number of such routes. Excess: total length above the shortest tours of the same customer sets. Belief err: believed minus true ex-post value. Ex-post: out-of-sample ex-post value.

Measure	Method	Runs[a]	Routes[b]	Excess	Belief err (%)	Ex-post
Max	Det-lower	0/25	0.00	0.0	-8.94	2221.0
Max	Det-free	0/25	0.00	0.0	-9.01	2232.4
Max	SORTIE-lower	0/25	0.00	0.0	-0.00	2174.8
Max	SORTIE-free	7/25	0.36	2.7	+0.00	2177.1
Range	Det-lower	0/25	0.00	0.0	-23.11	1099.7
Range	Det-free	23/25	1.00	87.8	-24.19	1088.5
Range	SORTIE-lower	0/25	0.00	0.0	+0.04	1098.4
Range	SORTIE-free	24/25	1.64	108.4	+0.06	1072.6
GMD	Det-lower	0/25	0.00	0.0	-19.46	437.7
GMD	Det-free	21/25	1.12	90.0	-20.02	425.9
GMD	SORTIE-lower	0/25	0.00	0.0	+0.07	434.5
GMD	SORTIE-free	23/25	1.52	82.4	+0.10	438.6

Table 5: E6: arc noise v, $n = 25$, $B = 1.10$, five instances per row.

v	Measure	Det belief err (%)	SORTIE-0 vs Det
0.1	GMD	-6.4	+1.05 (2/5)
0.1	Range	-8.7	-0.54 (1/5)
0.3	GMD	-22.0	+0.65 (2/5)
0.3	Range	-28.7	+0.96 (3/5)
0.5	GMD	-35.6	+1.84 (4/5)
0.5	Range	-42.7	+2.14 (5/5)

6.7 Larger instances (E7, E7b)

E7 runs $n \in \{50, 75, 100\}$ on six blocks each, with a 120 s limit, for the range, GMD and maximum (18 instance–measure pairs per size). With the screening lower level the search runs about 9000 iterations per run at $n = 50$ and 3500 at $n = 100$ for SORTIE-0 (Det: 15 000 and 4400). PBOI remains accurate (mean absolute error 0.14%, at most 0.9%) and the deterministic belief error is -17% on average, as on the small instances.

Table 6 pools the 18 pairs per size. SORTIE-0 improves on Det by about 2% at every size and is better on 12–13 of 18 pairs; the improvement is significant at $n = 50$ ($p = 0.035$) but not at $n = 75$ or $n = 100$ ($p = 0.17$). SORTIE-1 is significant at $n = 50$ but not beyond: its lower level runs a dichotomic search per route and completes only about 850 iterations at $n = 100$, and more than half of its runs were still improving in the second half of the time limit. Single runs at $n \geq 75$ are thus dominated by search noise, especially for the range and the GMD, whose value depends on a few routes.

E7b separates the method from the search noise at $n = 100$: the six instances and three measures are solved by Det, SAA and SORTIE-0 with three solver seeds each and a 240 s limit, and each method is represented by the mean ex-post value over its seeds (Table 7). Averaged over seeds, SORTIE-0 improves on Det by 4.2% (median 2.5%, better on 13 of 18 pairs, $p = 0.021$); SAA improves by 3.2% ($p = 0.10$). The difference between SORTIE-0 and SAA (1.1%, 12 of 18) is not significant ($p = 0.25$), and no single measure is significant on six instances. The seed-to-seed

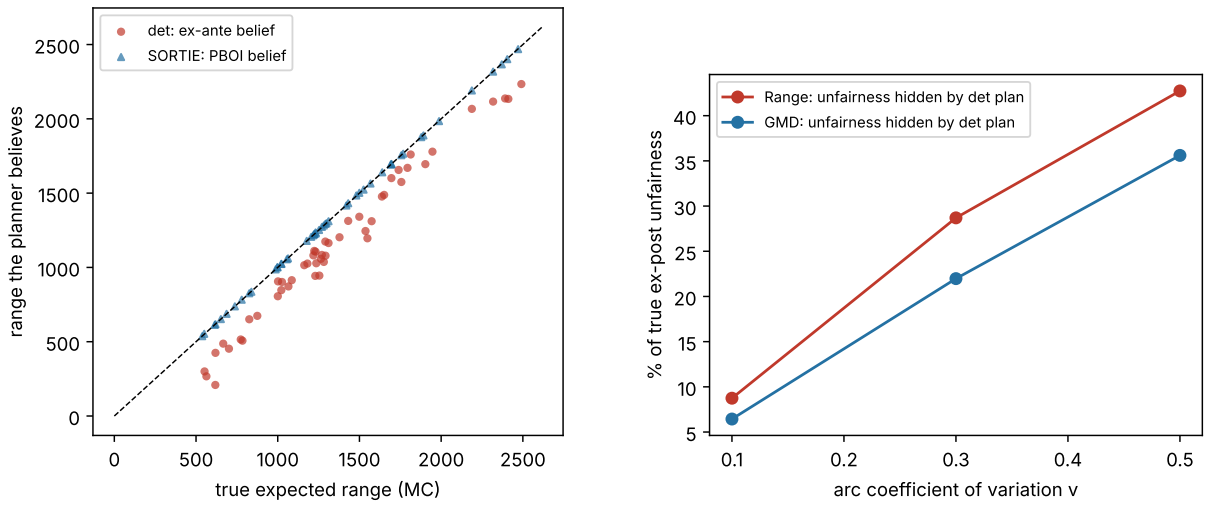


Figure 2: Left: range believed by Det (ex-ante) and by SORTIE (PBOI) against the true expected range, E3. Right: share of the true ex-post unfairness that a deterministic plan does not see, against v (E6).

Table 6: E7: improvement of the ex-post value over Det, $B = 1.10$, 120 s, pooled over six instances and three measures (range, GMD, max); format as Table 3.

n	SAA	SORTIE-0	SORTIE-1
50	+1.56 (12/18)	+2.04* (13/18)	+1.20* (15/18)
75	+1.96 (11/18)	+2.04 (13/18)	-0.29 (11/18)
100	+1.52 (10/18)	+2.21 (12/18)	-0.04 (8/18)

variation is itself informative: for the range, the ex-post value of the Det plan varies by 11% across seeds, that of SORTIE-0 by 3%. The deterministic range is flat over many plans whose realised fairness differs, and the search picks among them arbitrarily; the ex-post objective distinguishes them. Even at 240 s, 19% of the SORTIE-0 runs improved their in-model value by more than 0.5% in the second half of the run, so the absolute values at $n = 100$ are not fully converged.

6.8 Heterogeneous noise: congestion (E8)

E3 suggests that the stochastic refinement matters little when a route's variance is proportional to its length. E8 breaks that proportionality: arcs near a congestion centre have up to twice the base coefficient of variation and arcs far from it a third. We rerun the main grid ($B = 1.10$; range, GMD and max) on the congested versions of the same 25 instances, starting from the same cost plans.

The illusion becomes larger: a deterministic plan understates its range by 26% (against 18% under homogeneous noise), its GMD by 22% and its maximum by 10%, while the PBOI belief remains within 0.2% on average. The gain of optimising ex-post fairness, however, grows only slightly (Table 8). Pooled over 75 runs, SORTIE-0 improves on Det by 1.51% (median 1.24%; better in 53, worse in 3; $p < 10^{-8}$) and SORTIE-1 by 1.66% (better in 57, worse in 10), against 1.36% for SORTIE-0 on the same cells without congestion. By measure the change is 0.94% → 1.40% for the GMD, 2.08% → 2.15% for the maximum and 1.07% → 0.96% for the range. SAA gains 1.28%, but with a median of only 0.47%, and SORTIE-0 beats it head to head in 49 of 75 runs (worse in

Table 7: E7b, $n = 100$, $B = 1.10$, 240 s, three solver seeds: improvement of the seed-averaged ex-post value over Det (six instances per measure; format as Table 3) and coefficient of variation of the ex-post value across seeds, averaged over instances.

	vs Det, mean of 3 seeds		Seed CV (%)		
Measure	SAA	SORTIE-0	Det	SAA	SORTIE-0
Range	+6.45 (4/6)	+8.68 (5/6)	11.3	4.7	3.0
GMD	+2.49 (3/6)	+3.18 (4/6)	6.4	2.7	4.5
Max	+0.78 (4/6)	+0.86 (4/6)	1.4	0.8	1.4
All	+3.24 (11/18)	+4.24* (13/18)	SORTIE-0 vs SAA: +1.09 (12/18)		

Table 8: E8: congested arc noise, $B = 1.10$, 25 instances per row; improvement of the ex-post value over Det (format as Table 3).

Measure	SAA	SORTIE-0	SORTIE-1
GMD	+0.76 (11/25)	+1.40* (13/25)	+1.82* (15/25)
Max	+1.87* (23/25)	+2.15* (23/25)	+2.21* (24/25)
Range	+1.22 (13/25)	+0.96* (17/25)	+0.94* (18/25)

6; $p < 10^{-6}$). The $\mathrm{CVaR}_{0.9}$ of the range is 1919 for Det, 1905 for SAA, 1912 for SORTIE-0 and 1907 for SORTIE-1. Heterogeneous noise thus makes a deterministic plan's self-assessment much worse, but in our instances it changes the best plan only a little more than homogeneous noise does: the congested arcs are concentrated in one part of the network, and the assignment that balances lengths still roughly balances risk there.

7 Conclusions

Fair routing under random travel times has to say whose fairness it measures. Drivers experience realised durations, so the relevant quantity is the expected unfairness of the realised day. A deterministic fair plan cannot see this quantity, and in our experiments it understated it by 8% (maximum) to 27% (variance) at $v = 0.3$, and by up to 43% at $v = 0.5$. A multiplicative common shock, by contrast, leaves the expected value of every ordered measure unchanged (Proposition 2), although it does inflate the variance measure and the tail of unfairness.

The Poisson–binomial order-statistic integral evaluates any ordered measure of independent route durations exactly, in under a millisecond for five vehicles, and its derivatives, the rank prices, are both interpretable and useful: they identify the routes a fairness objective would like to lengthen, and screening moves with them gave plans 6–9% better on average than random screening. Under uncertainty, route consistency no longer follows from monotonicity, so it has to be part of the model; the mean–standard-deviation lower level makes it computable by parametric Held–Karp. Without a lower level, both the deterministic and the stochastic planners produced detours in almost every range-optimised plan.

Optimising ex-post fairness improved the realised fairness by about one percent over a deterministic fair plan under homogeneous noise (up to 2% for the maximum), consistently but modestly, because balancing route lengths already roughly balances route risks when variance grows with length. When noise is heterogeneous, as with congestion concentrated in part of the network, the illusion grows to 26% for the range, but the gain of SORTIE grows only to about 1.5%. The main practical value of the stochastic model in our experiments is therefore that it tells the planner,

accurately, how unfair a plan will be, and that it guarantees consistent tours; the improvement of the plan itself is real and significant but small. On instances with 100 customers the faster search makes the comparison with the deterministic plan conclusive when search noise is averaged over seeds (a 4% improvement, $p = 0.02$), but it does not separate SORTIE from a 200-scenario SAA at that size.

Limitations. The lower level is exact only for routes of up to 15 customers; beyond that it relies on multi-start local search (warm-started from the screening tour), and the bilevel constraint is then satisfied only approximately. PBOI assumes conditionally independent route durations; dependence beyond a common multiplicative factor (for example spatially correlated congestion affecting several routes) would need the copula or simulation layer we did not develop. At $n = 100$ single runs are still noisy and some are not converged after 240 s; the risk-consistent variant ($\kappa = 1$) is too slow to converge at $n \geq 75$; and at that size SORTIE could not be separated statistically from a 200-scenario SAA. The time limits are per run and are slightly exceeded by the final iteration. All instances are slices of one benchmark file with five vehicles.

Future work. Exact methods for the stochastic bilevel problem (branch-and-cut with PBOI-based bounds), correlated route durations, multi-day fairness, in which unfairness can be compensated across days, and time-dependent travel times.

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